Preston & Dietz
The Art of Experimental Physics.
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pg 303 – 315

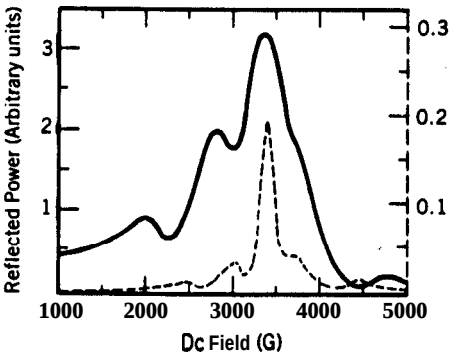


FIGURE 16.16 Spectrum reported in reference 4 for B_0 perpendicular to the (111) face of the crystal. The scales for the solid line curve (left) and dashed line curve (right) refer to the same input power.

conservation of energy (see Exercise 5) to calculate D and $g_{\parallel}$. Compare your values with those given in Exercise 5. Also measure the linewidth of the most distinct of the three lines and calculate T_2.

17. ELECTRICAL CONDUCTMTY AND THE HALL EFFECT

Historical Note

The 1985 Nobel prize in Physics was awarded to
Klaus von Klitzing, Germany
For his discovery of the quantized Hall resistance.

APPARATUS [Optional Equipment in Brackets]

Cryostat
[Oven]
Laboratory electromagnet with power supply and gaussmeter
Rectangular semiconducting crystal (e.g., a low-impurity piece of germanium or other semiconductor with dimensions of about 1 x 2 x 10 mm and $\rho > 0.5$ Ω-cm at 300 K)
Electrometer or potentiometer for measuring Hall voltages
Ammeter
Voltmeter
4-V battery
Assorted resistors
Copper-constantan thermocouple with potentiometer

OBJECTIVES

To measure the Hall voltage and conductivity for a semiconductor as functions of temperature and to interpret these data qualitatively.

To derive the sign of the extrinsic charge carriers, the Hall mobility, and the energy gap from the data.

To observe and understand qualitatively the phenomenon of magnetoresistance.

To understand the basic principles underlying the Hall effect and its relationship to charge transport mechanisms in semiconductors.

Key CONCEPTS

Hall voltage
Hall resistance
Conductivity
Fermi energy
Valence/conduction band
Effective mass
n-type/p-type semiconductor
Thermomagnetic/thermoelectric effects
Hall coefficient
Free electron model
Mobility
Energy gap
Holes
Intrinsic/extrinsic carriers
Magnetoresistance

REFERENCES

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2. C. Kittel and H. Kroemer, ThermalPhysics, 2d ed., Freeman, San Francisco, 1980. Chapter l3 is a thorough discussion of semiconductor statistics which includes calculations of carrier concentrations.

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7. E. H. Putley, The Hall Effect and Semi-conductor Physics, Dover, New York, 1968. Discusses theoretical and experimental aspects of charge transport processes in semiconductors. Includes band theory, Hall and magnetoresistance effects, and thermal effects.

8. B. I. Halperin, Sci. Am. 254, 52 (1986). A descriptive account of the integral and fractional quantized Hall effects.

9. Experiment 6, introduction to Computer-Assisted Experimentation, In this experiment the Hall sample is interfaced to the computer, and the computer is used to measure the Hall voltage as a function of both the magnetic field and the applied current.

INTRODUCTION

If a block of conducting material that carries a current I along the x direction is placed in a magnetic field $B = B\hat{z}$, as in Figure 17.1, an electrostatic potential difference, referred to as the Hall voltage V_H, is produced between faces 3 and 4 along the y direction. The production of this voltage, first observed by E. H. Hall in 1879, is called the Hall effect and is due to the deflection along the y axis of the moving charge carriers in the material. The ratio V_H/I, called the Hall resistance, is observed to increase with increasing B.

The Hall effect involves several independent variables and so is the operating principle for a rather wide variety of practical instrumentation. For the experimental physicist, the Hall-effect gaussmeter, which provides convenient and accurate measurements of laboratory magnetic fields. is perhaps the most well known of these devices. Other applications include, but are by no means limited to, displacement indicators, tachometers, ammeters, modulators, D/A converters, multipliers, and spectrum analyzers.

There are two regions in which experimental studies of the Hall effect yield useful results. For low fields, defined by the condition $\omega_c\tau \ll 1$ ($\omega_c \equiv eB/m$ is the cyclotron frequency and τ is the mean time between collisions for the charge carriers), scattering processes are of importance in determining the motion of charge carriers. In this region, the behavior of the Hall voltage, taken together with electrical conductivity measurements, is a rich source of information regarding charge transport properties of metals and semiconductors. Concentrations, mobilities, and signs of charge carriers can be determined under the low-field conditions, which are appropriate to all of the measurements described in this experiment.

For the high-field region, defined by $\omega_c\tau \gg$ I, the motion of the charge carriers is determined by the externally applied fields and the behavior of the Hall voltage can yield information about the quantum-mechanical band structure of the material under investigation. If, additionally, the temperature is close enough to absolute zero, the effect of the magnetic field on the charge carrier states is manifested in the quantum behavior of the Hall resistance. In 1985, Klaus von Klitzing won the Nobel prize in physics for his observation of the integral quantized Hall effect, in which the Hall resistance for a two-dimensional system does not increase linearly with B as it does in the high-field region, but rather exhibits plateaus as B increases. Electrical resistance vanishes for values of B for which the Hall resistance exhibits a plateau; the reciprocals of the values of the Hall resistance (the Hall conductances) on these plateaus are given with great precision as integer multiples of e^2/h, thus defining a basic unit of conductance. Such measurements can establish precise

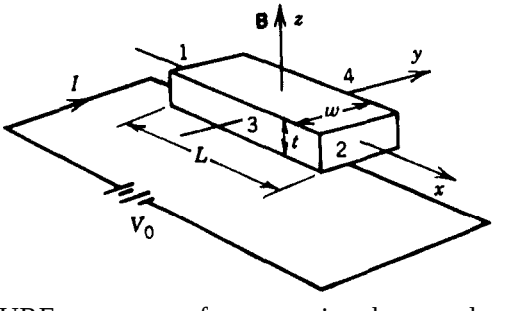


FIGURE 17.1 Geometry for measuring the Hall voltage for a conducting sample in a magnetic field.

standards of resistance, provide precise values for fundamental constants, and lead to more stringent tests of quantum electrodynamics.

Electrical Conductivity

For simple metals (e.g., the alkalis), many properties, including charge transport, are understood quite well in terms of the free electron model. In this picture, current is carried by electrons (charge -e and number density n) that are accelerated by an applied electric field E as they undergo collisions with lattice imperfections, impurities, and phonons (thermal vibrations of the lattice). The current density $J = nev$ can be calculated by noting that if the mean time between collisions is τ, then the effect of collisions can be approximated as a retarding force of magnitude mv/τ, where v is the average drift velocity acquired by the electrons between collisions. Newton's second law for these electrons reads, in this approximation

$$m\left(\frac{d}{dt}+\frac{1}{\tau}\right)\mathbf{v} = -e\mathbf{E} \qquad (\mathrm{N}) \tag{1}$$

In the steady state the retarding force just balances the force due to **E**, so that the time derivative vanishes, yielding

$$\mathbf{v} = -\frac{e\tau}{m}\mathbf{E} \qquad (\mathrm{m/s}) \tag{2}$$

The electron density, n, is determined only by the density of atoms in the crystal and the number of valence electrons per atom. The current density produced by E is thus given by

$$\mathbf{J} = \frac{ne^2\tau}{m}\mathbf{E} \qquad (\mathrm{A/m^2}) \tag{3}$$

and the conductivity σ, defined by $J = \sigma\mathbf{E}$, is just

$$\sigma = \frac{ne^2\tau}{m} \qquad (\Omega \cdot \mathrm{m})^{-1} \tag{4}$$

The result is also conveniently expressed in terms of the mobility μ of the charge carriers, defined by

$$\mu \equiv \frac{|\mathbf{v}|}{|\mathbf{E}|} \qquad (\mathrm{m^2(V \cdot s)^{-1}}) \tag{5}$$

In terms of μ, the conductivity for a metal is obtained from equations 2 and 3:

$$\sigma = ne\mu \qquad ((\Omega \cdot \mathrm{m})^{-1}) \tag{6}$$

In the quantum-mechanical description of a metal, the electrons are arranged in energy bands, as depicted in Figure 17.2*a*. Electrons occupy states within the allowed bands up to the Fermi energy $\mathscr{E}_F$. In a metal, $\mathscr{E}_F$ falls below the top of a band, so that an electric field can accelerate electrons by promoting them to unoccupied states of higher energy within this same band.

In a semiconducting material the situation is quite different. Examples of semiconductors are the crystalline forms of elements from group IV of the periodic table (e.g., Ge or Si) or the compounds of elements from groups III and V (e.g., GaAs or AISb). In the pure forms

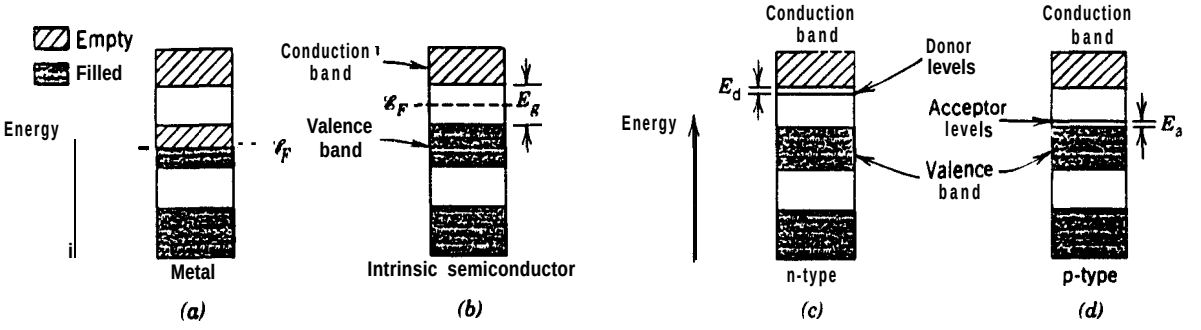


FIGURE 17.2 (a) Electronic energy bands for a metal. (b) Electronic energy bands for an intrinsic semiconductor (c) Electronic energy bands for an n-type semiconductor. (d) Electronic energy bands for a p-type semiconductor.

of these materials at $T = 0$ K, there are no unoccupied states near the top of the electron energy distribution; that is, the **valence band** is filled, as indicated by Figure 17.26. The unoccupied states of lowest energy are located in the **conduction band** at an energy E_g (the width of the *energy gap*) above the top of the valence band. Thus, an applied electric field cannot accelerate any of the electrons, and no current can flow. As T is raised above 0 K, some of the electrons near the top of the valence band are thermally excited into the conduction band, where they behave as free carriers of negative charge. The vacancies in the valence band left behind by these excited electrons are known as holes and behave as free carriers of positive charge. These thermally produced charge carriers are referred to as **intrinsic** carriers. The density of thermally excited electrons (n_e) and holes (n_h) will, along with their respective mobilities, determine the conductivity of the material. For a pure semiconductor, the intrinsic carrier densities are calculated from considerations of statistical mechanics (see reference I or 2):

$$n_e = n_h = 2\left(\frac{k_B T}{2\pi\hbar^2}\right)^{3/2} (m_e m_h)^{3/4}\, e^{-E_g/2k_B T} \qquad (\mathrm{m}^{-3}) \tag{7}$$

where k_B is **Boltzmann's** constant and m_e and m_h are the *density-of-states effective masses* of the electrons and holes in the crystal as determined from the band structure.

EXERCISE I

Why are the hole and electron densities equal in a "pure" semiconductor?

Semiconductors are never pure. The presence of impurities, whether intended or not, markedly affects the carrier concentration and, hence, the conductivity of the material. Addition of group V atoms, which have an "extra" valence electron, creates an n-type semiconductor in which there are occupied donor levels located at an energy E_d below the conduction band, as illustrated by Figure 17.2*c*. Electrons occupying these levels are easily excited into the conduction band at temperatures much lower than that required to produce intrinsic carriers. On the other hand, a p-type semiconductor results from the addition of group III impurities. These atoms, which have one less valence electron than the atoms of the host crystal, create vacant acceptor levels located at an energy E_a above the valence band, as illustrated by Figure 17.2*d*. Electrons from the top of the valence band can readily be excited into these levels, leaving holes in the valence band to act as free charge carriers. Charge carriers created in this way are termed **extrinsic,** or impurity carriers. The calculation of the extrinsic carrier densities is somewhat involved, but if only one type of impurity

is present, its associated carrier density n_{ex} can be expressed in closed form. If we neglect the concentration of intrinsic carriers, then

$$n_{ex} = \frac{n^*}{4}\left\{\sqrt{1+\frac{8n_I}{n^*}} - 1\right\} \quad (m^{-3}) \tag{8}$$

In this expression the parameter n^* is defined by

$$n^* \equiv 2\left(\frac{m^* k_B T}{2\pi\hbar^2}\right)^{3/2} e^{-E_I/K_B T},$$

n, is the impurity atom concentration, and we set $E_I = E_d$, m* = m_e for n-type materials, and $E_I = E_a$, m* = m_h for p-type materials. The energy E_I is called the **impurity ionization energy.**

EXERCISE 2

Show that in the low-temperature limit, $k_B T \ll E_I$, the concentration of extrinsic carriers has a temperature dependence given by

$$n_{ex} \propto T^{3/4} e^{-E_I/2k_B T} \quad (m^{-3}) \tag{8'}$$

Note that this expression is good only in the temperature range indicated, and that as T is raised much beyond $E_I/2k_B$, the extrinsic carrier concentration saturates so that $n_{ex} \cong n_I$. For still higher temperatures, in the intrinsic region, intrinsic carrier production dominates charge transport processes in accordance with equation 7.

Carrier mobilities $\mu_e \equiv |\mathbf{v}_e|/|\mathbf{E}|$ and $\mu_h \equiv |\mathbf{v}_h|/|\mathbf{E}|$ for electrons and holes, respectively, are defined by analogy with the definition for metals stated in equation 5. These quantities are related, by comparison with equation 2, to the mean times between collisions for electrons and holes, τ_e and τ_h, by the relations

$$\mu_e = \frac{e\tau_e}{m_e} \qquad \mu_h = \frac{e\tau_h}{m_h} \quad (m^2(V\cdot s)^{-1}) \tag{9}$$

Mobilities are limited by scattering processes involving impurities and thermal phonons. in the temperature range where phonon scattering is of primary importance, carrier mobilities are expected to vary as $T^{-3/2}$. Hole mobihties are typically smaller than those for electrons because the distribution of the valence states allows a greater variety of scattering mechanisms.

In a semiconductor, both types of charge carriers contribute to the conduction of current. Following the same reasoning as for equations l-6 above, we extend the expression for the conductivity to include the case in which both electrons and holes may be present:

$$\sigma = n_e e\mu_e + n_h e\mu_h \quad ((\Omega\cdot m)^{-1}) \tag{10}$$

Hall Effect

For a metal or for an n- or p-type semiconductor in the extrinsic temperature range, electrical conduction is due to a single carrier type with charge q and mobility μ. If a magnetic field B is applied along $\hat{z}$, as in Figure 17.1, then a carrier of charge q experiences a Lorentz force F = q(E + v x B). If the definition of carrier mobility is expanded to include magnetic forces, then the drift velocity is given by

$$\mathbf{v} = \pm\mu(\mathbf{E} + \mathbf{v}\times\mathbf{B}) \quad (m/s) \tag{11}$$

where the plus sign is used for holes, and the minus sign is used for electrons. For this configuration of fields, the drift velocity components are related by

$$v_x = \pm \mu(E_x + v_y B_z) \qquad v_y = \pm \mu(E_y - v_x B_z) \qquad v_z = \pm \mu E_z \qquad \text{(m/s)} \qquad (12)$$

The components of the current density $\mathbf{J} = nq\mathbf{v}$ are thus

$$J_x = \pm \mu(nqE_x + J_y B_z) \qquad J_y = \pm \mu(nqE_y - J_x B_z) \qquad J_z = \pm \mu nq E_z \qquad \text{(A/m}^2\text{)} \quad (13)$$

If the current is being supplied by a battery attached to faces I and 2 of the sample, then in the steady state we must have $J_y = J_z = 0$. This implies that there is a component of E transverse to the current flow:

$$E_y = \frac{J_x B_z}{nq} \qquad \text{(V/m)} \qquad (14)$$

This is accounted for physically by the buildup of a static charge distribution on faces 3 and 4 which produces an electric field along the y axis that compensates for the force of **B** on the carriers. The **Hall coefficient** R_H is defined by

$$R_H = \frac{E_y}{J_x B_z} \qquad \text{(m}^3\text{/C)} \qquad (15)$$

We can restate this definition in terms of experimentally measured quantities by referring to Figure 17.1 and noting that the Hall voltage $V_H = V_3 - V_4$ is expressed in terms of E_y *as* $V_H = wE_y$, and that $J_x = I/wt$. Thus, the experimental Hall coefficient is expressed in terms of measurable quantities as

$$R_H = \frac{V_H t}{IB_z} \qquad \text{(m}^3\text{/C)} \qquad (16)$$

By comparing equations 14 and 15, it can be seen that the theoretical Hall coefficient for the case of a single carrier type is expressed in terms of the carrier density as

$$R_H = \frac{1}{nq} \qquad \text{(m}^3\text{/C)} \qquad (17)$$

Note that the carrier density in this case can be calculated directly from a measurement of R_H, and that the sign of R_H can be used to determine the sign of the charge carriers. If both R_H and σ are measured, then the **Hall mobility** μ_H is determined by

$$\mu_H = |R_H|\sigma \qquad \text{(m}^2\text{(V} \cdot \text{s)}^{-1}\text{)} \qquad (18)$$

EXERCISE 3

Using Figure 17.1 along with the definition of equation 15 and the right-hand rule for the Lorenz force, verify that R_H is positive for extrinsic p-type semiconductors.

In semiconductors where both electrons and holes contribute to conduction, the expression for R_H involves the densities and mobiiities of both carriers. The calculation of R_H for this case is a straightforward extension of the one above for a single species of carrier. The

current density is the sum of the electron and hole current densities, and thus has a y component that is

$$J_y = n_h e v_{hy} - n_e e v_{ey} \qquad (\mathrm{A/m^2}) \tag{19}$$

where v_{hy} and v_{ey} are y components of the hole and electron velocities, respectively. Substituting v_y given in equation 12 for each carrier into this expression for J_y yields

$$J_y = (n_h \mu_h + n_e \mu_e) e E_y + (n_e \mu_e^2 - n_h \mu_h^2) e B_z E_x \qquad (\mathrm{A/m^2}) \tag{20}$$

Here we assume that we are working in the low-field region, as discussed in the introduction, so that we can neglect terms containing B^2 in comparison with those containing B. As for the single carrier calculation, we demand $J_y = 0$ in the steady state. Writing $J_x = \sigma E_x$, with σ given approximately by equation IO, gives us for R_H

$$R_H \equiv \frac{E_y}{J_x B_z} = \frac{n_h \mu_h^2 - n_e \mu_e^2}{e(n_e \mu_e + n_h \mu_h)^2} \qquad (\mathrm{m^3/C}) \tag{21}$$

EXERCISE 4

In light of the observation that hole mobilities are typically much smaller than electron mobilities, what is the expected sign of the Hall coefficient for an intrinsic semiconductor? What can you conclude about the impurities in a semiconductor for which R_H changes sign as the temperature is raised?

Note that for a material with a single type of charge carrier with a well-defined mobility, the steady-state condition $J_y = 0$ is equivalent to requiring that the y component of the Lorentz force on the carriers be zero, so that the carrier trajectories are straight lines along the x axis. In this instance it is expected that the sample in Figure 17.1 would have a measured resistance between faces 1and 2 that is not affected by the applied magnetic field B. If, however, the carriers have a distribution of mobilities (e.g., in the two-carrier case), then $J_y = 0$ is not equivalent to zero Lorentz force for all carriers, and some will develop drift velocities with nonzero y components. This will result in curved carrier trajectories and an accompanying increase in the measured resistance of the sample. The resistance measured in the presence of B is termed the **magnetoresistance,** and is important for the determination of electronic structures. This same mechanism explains why R_H has a slight dependence on B.

EXPERIMENT

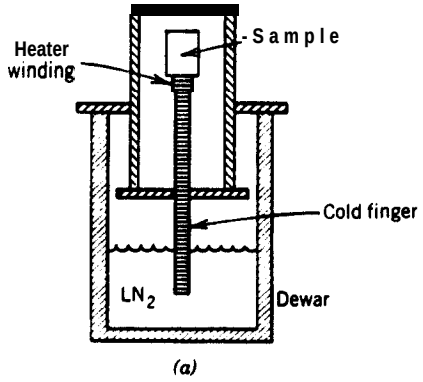
Cryogenics

Exploration of the temperature behavior of the Hall coefficient and conductivity of a sample requires a means of measuring and slowly varying its temperature as voltage and current measurements are made. Aside from the use of commercially available cryostats, there are several convenient methods for accomplishing this which vary in cost, sophistication, and range of temperatures available. Some that are suitable for this experiment are suggested below.

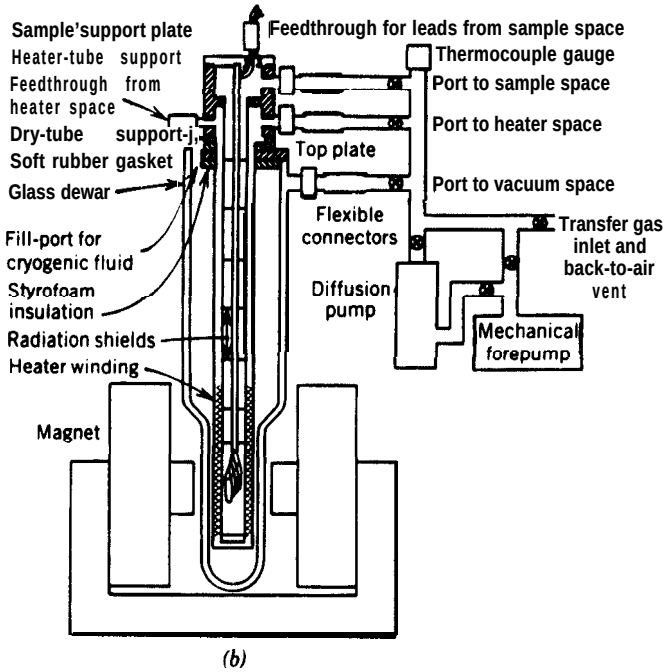
1. A mixture of dry ice and alcohol used as a cryogenic fluid allows exploration of temperatures from about 190 K to around room temperature. The alcohol should be added to a Dewar containing the dry ice and, after the mixture has reached a fairly

quiescent state, the alcohol should be poured off (with a minimum of dry ice) into a second Dewar or beaker in which thermal contact is made with the sample. Measurements should be taken as the temperature rises. When the rate of temperature increase becomes inconveniently slow, higher temperature data may be taken with the use of an oven.

2. The apparatus shown schematically in Figure 17.3*a* allows measurements down to about 80 K by utilizing a thermally conducting rod (e.g., of copper) as a "cold finger,"* which thermally links the sample and a liquid nitrogen (LN_2) bath. The overall efficiency of the link may be optimized by replacing a portion of the rod with material of lower thermal conductivity. Operation of the heating coil when liquid nitrogen is in the Dewar allows variation of the sample temperature. As in method 1, measurements can be taken "on the fly" as the temperature rises rather than awaiting stabilization at each point.



(a)



(b)

FIGURE 17.3 (*a*) Cold finger cryostat, (*b*) Low-cost cryostat described by Martin and Sidles, reference 4.

3. The construction of the cryostat shown in Figure 17.36 is discussed in detail by Martin and Sidles (reference 4) as a low-cost option for the instructional laboratory. The apparatus shown is suitable for operation down to liquid nitrogen temperature, but is readily modified for operation with liquid helium.

In all of the above systems, a copper-constantan thermocouple in thermal contact with the sample may be used for temperature measurements. One of the junctions should be maintained in an ice-water reference bath at 0 °C so that the calibration tables (reference 5) may be used. If desired, a linear correction may be made to these tables based on the thermocouple emf measured when the junction is dipped in a cryogenic liquid of known temperature (e.g., liquid netrogen at 77 K). For the temperature range available in method 1, a low-temperature toluol thermometer may be used instead of a thermocouple.

Electrical Connections

Electrical leads may be soldered to the sample, but care must be taken to wet the sample with the solder and to avoid rectifying contacts. A convenient and effective alternative to soldering is the use of the sample holder described in reference 4 and shown in Figure 17.4, where part a of the figure is a front view of the holder and part b is a cross-sectional side view. Contacts for measuring Hall potentials are made with segments of 20-mil tungsten wire, the ends of which are sharpened by anodic oxidation in a potassium hydroxide solution with a graphite cathode. Faces of the electrodes making contact with the ends of the sample are coated with indium solder; making the end faces equipotentials in this way will not significantly "short out" the Hall voltage measured across the middle as long as

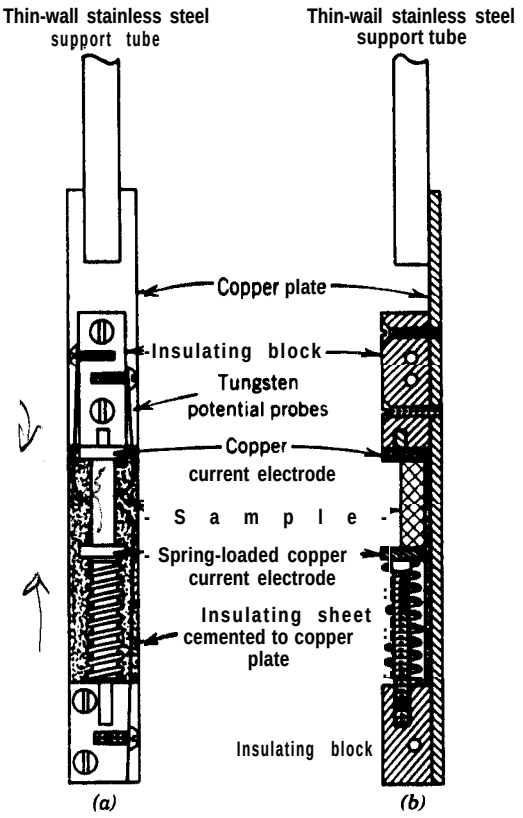


FIGURE 17.4 (*a*) Sample holder (front view). (*b*) Sample holder (cross-sectional side view).

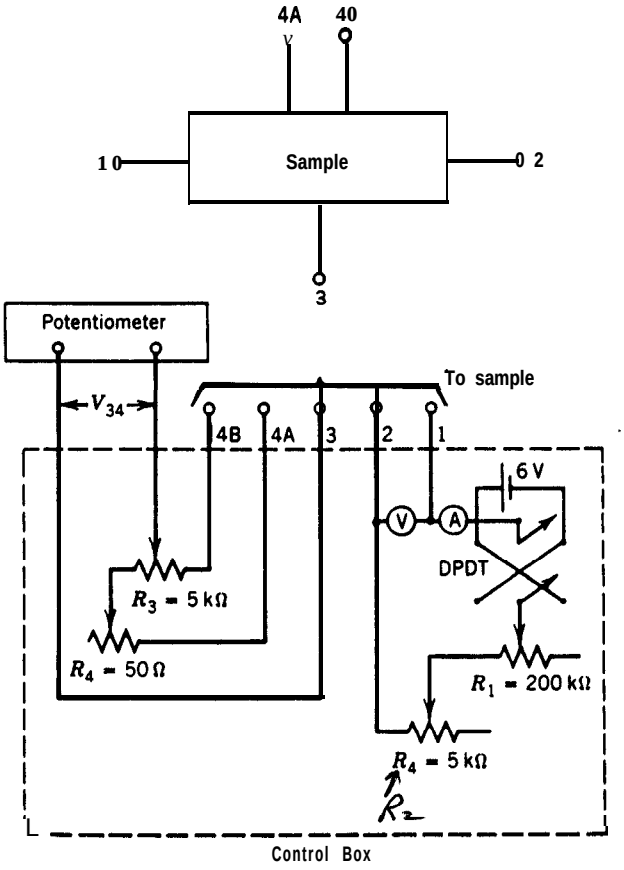


FIGURE 17.5 Circuit for conductivity and Hall voltage measurements.

$L/w > 4$ for the sample. Spring loading of one current electrode eliminates difficulties caused by differential thermal expansion and contraction. The base of the holder is a thin copper plate to which the thermocouple may be attached and which is in thermal contact with but electrically insulated from the sample and electrode supports. The surface of the sample should be cleaned of grease and oxides before insertion into the holder; after insertion an ohmmeter may be used to verify that nonrectifying contacts have been made.

The circuit for the conductivity and Hall voltage measurements is shown in Figure 17.5. Connections from the control box to the sample may be made conveniently with a Jones strip. Resistances R_1 and R_2 control the current through the sample. If a single probe were used on each side of the sample to measure V_H, there would be an "IR drop" contribution to the measured Hall voltage if the probes were not precisely located along a line perpendicular to the direction of current flow. To eliminate this effect, two probes, 4A and 4B, are used, and adjustment of resistances R_3 and R_4 is equivalent to sliding a single probe between these two contact points. Thus, these resistances should be adjusted at the beginning of the measurements with the sample current flowing and with $B = 0$ until $V_{34} = 0$.

Because of several thermomagnetic effects, the measured voltage V_{34} may not be the "true" Hall voltage V_H. Reference 3 gives a concise description of the Nemst effect, the Righi-Leduc effect, and the Ettingshausen effect, along with suggestions for reducing their respective contributions to measured voltages. The Ettingshausen effect cannot be separated from the Hall effect in the present experiment, but it will be small in samples with good bulk thermal conduction, The Nemst and Righi-Leduc effects, along with any residual IR drop,

can be compensated for by averaging V_{34} over both directions of current and field. Thus, the Hall voltage corresponding to a current I and field B should be taken as an average of four separate measurements:

$$V_H = \tfrac{1}{4}[V_{34}(I, B) - V_{34}(I, -B) - V_{34}(-I, B) + V_{34}(-I, -B)] \quad \text{(V)} \qquad (22)$$

Thermoelectric effects can also affect V_{12}, the voltage measured between points 1 and 2 on the sample, if there are temperature gradients. Thus, the conductivity should be calculated using an average voltage V_0, given by

$$V_0 = \tfrac{1}{2}[V_{12}(I) - V_{12}(-I)] \quad \text{(V)} \qquad (23)$$

Measurements

Measure the dimensions of the sample. With the sample removed from the pole pieces of the magnet, zero the IR drop as discussed above. Place the sample between the pole pieces and, with a current of a few milliamperes flowing in the sample, rotate it until V_H is maximized; this procedure aligns the apparatus for future measurements. Note how the directions of B and I in the sample correlate with the polarity of the meter readings for the magnet and sample current.

With the sample at room temperature, select a convenient sample current and obtain a few values of V_H as a function of B at constant I. Do the same for V_H at constant B as a function of 1.

EXERCISE 5

Do plots of V_H against B and V_H against I verify the linear relationships implied by equation 16?

Bring the sample to the lowest temperature in the range to be investigated and set the sample current at a convenient value, to be maintained throughout the experiment. Take measurements every few degrees as the temperature rises, reading the thermocouple emf both before and after the voltage measurements at each temperature. For each temperature do the following:

(a) With the magnetic field at a constant magnitude of about 1kG, record V_{12} and V_{34} for current flow in both directions.
(b) Record V_{12} with $B = 0$, for both current directions.
(c) With the magnetic field reversed but equal in magnitude to that in (a), record V_{34} for both current directions.

From measurements of V_{12}, V_{34}, and I, determine the Hall coefficient and electrical conductivity for each temperature.

EXERCISE 6

Plot both R_H and σ against T and explain the behavior of these curves. Can you identify an extrinsic temperature region? An intrinsic temperature region?

EXERCISE 7

Plot $\ln(R_H T^{3/2})$ against I/T. From the linear portion of this curve in the intrinsic region, determine E_g, the energy gap in the semiconductor. Assume for this calculation that $n_e = n_h$ in this temperature range. What assumption(s) do you need to make about the carrier mobilities? How does your value for E_g compare with that given by reference 1? Bear in mind that E_g is a function of temperature.

EXERCISE 8

Estimate the impurity ionization energy E_i from the behavior of the low-temperature portion of the curve plotted in response to Exercise 7, if possible. What is the sign of the impurity charge carrier?

EXERCISE 9

Determine the energy gap arid, if possible, the impurity ionization energy from the conductivity data. What assumptions are you making? How do these values compare with those obtained from the Hall coefficient data?

EXERCISE 10

Calculate the Hall mobility μ_H given by equation 18 for each temperature. From a plot of In μ_H versus In T, can you identify a temperature range in which a simple power law fits the data? If such a range exists, does the exponent in the power law suggest lattice scattering (discussed following equation 9) as the primary limiting factor for the mobility? It should be noted in this connection that the Hall mobility, defined experimentally by equation 18, is in general not equal to the mobility derived from conductivity measurements alone. Why might this be so?

EXERCISE 11

Calculate, for each temperature, the ratio $(R_m - R)/R$, where R_m is the resistance measured in the presence of a magnetic field and R is the resistance in zero magnetic field. Can you explain qualitatively the behavior of this ratio as the temperature is increased7